DHS/MIS malaria prevalence and child mortality analysis

Main results

- **Higher malaria prevalence is associated with higher post-neonatal mortality, and this holds regardless of covariate adjustment.** Each 10 percentage-point increase in PfPR2-10 is associated with a 6.9% increase in post-neonatal mortality in the linear summary model (95% CI 4.5%–9.4%; `p<0.001`), and with a 9.0% increase (6.4%–11.8%) at the reference year 2013 in the selected model, which lets the slope change over calendar time. The estimate is essentially unchanged whether we use the regional covariate block alone or the full regional-plus-national block that includes national vaccine coverage and child HIV prevalence.
- **The primary analysis now uses a 12-month mortality window rather than the DHS default of 60 months.** The exposure is MAP prevalence in the survey year, so a 60-month window centres the outcome about 2.5 years before the exposure it is regressed on; a 12-month window nearly removes that mismatch. The cost is precision (the interval on a regional rate roughly doubles) and 56 region-years that record no post-neonatal or no neonatal death within the year, leaving **1,055 region-years (118 surveys, 35 countries)**. The dose-response barely moves across windows once the model structure is held fixed.
- **Neonatal mortality, the negative control, is null.** The linear neonatal estimate is +0.1% per 10 PfPR points (95% CI −2.2% to +2.5%; `p=0.92`), the prespecified structure refitted to neonatal mortality gives +1.2% (−1.4% to +3.8%; `p=0.38`), and the Bayesian prevalence-by-time surface fitted to neonatal deaths sits on zero at every year and in all nine subgroups. At 60 months on the same panel the control had drifted to +1.7% (`p=0.045`); the shorter window removes that. This, with the stability under HIV and vaccine adjustment, reduces the concern that the main result reflects broad differences in child survival between higher- and lower-prevalence settings.
- **The data support letting the prevalence effect change over time, but do not say how.** By AIC, three structures tie within 0.44 points: a linear PfPR term with a time interaction, a spline with a `ti` interaction, and the additive spline `s(PfPR)+s(year)`. Bayesian refits with shared priors cannot be separated by pointwise leave-one-out (differences of 0.2–0.4 against standard errors of 1.2–2.3). Holding out whole surveys in a grouped 10-fold cross-validation does separate them: the additive model trails the full tensor surface by 7.1 in held-out log predictive density (pointwise SE 3.4, survey-level SE 4.4) and the linear-in-time model trails it by only 1.1 (SE 1.7). The gain sits in survey regions with prevalence of 30% or more. The attributable fractions at 2013 are nearly identical under all three structures; what differs is the historical trend.
- **Population-average malaria-attributable fractions of post-neonatal mortality are 7.5%, 22.2% and 34.6% at 10%, 30% and 50% PfPR2-10** (versus a 1% counterfactual, reference year 2013, selected model). The Bayesian surface, which propagates uncertainty in how smooth the curve is, gives 11.6%, 27.8% and 36.1% with wider intervals (2.6%–21.4%, 15.1%–39.2%, 22.9%–47.4%).
- **The best-fitting national extrapolation estimates about 527,000 malaria-attributable post-neonatal deaths in 2024**, 95% interval 373,000–658,000 (spline with `ti` interaction); the full `te` surface gives 503,000 and a time-stable spline 435,000. IHME/GBD's 2024 under-five figure is about 428,000 and the WHO under-five proxy about 434,000, so the point estimate is 23% above IHME with overlapping intervals. This is not a validation against a common estimand: the model captures direct and indirect malaria-associated mortality, whereas IHME and WHO assign deaths specifically to malaria. The 2000–2024 trend remains unidentified: the two best structures, 0.3 AIC apart, imply changes of −5% and −53%.

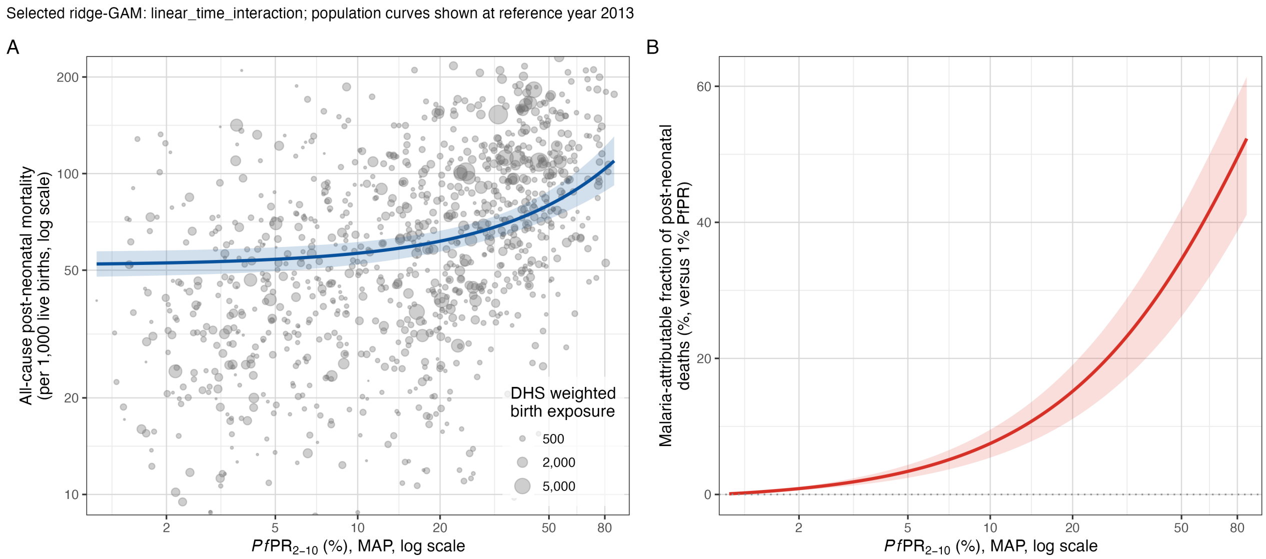

Figure 1: Malaria and post-neonatal mortality. **A:** all-cause post-neonatal mortality versus PfPR2-10, with point size proportional to the survey region's weighted birth exposure and the fitted ridge-GAM curve. **B:** the model-estimated malaria-attributable fraction of post-neonatal deaths as a function of PfPR2-10.

The sections below give the data, methods and supporting detail.

Data and methods

Sample. The unit of analysis is a DHS/MIS survey-region-year (Births Recodes, sub-Saharan Africa, 2000–2024; AIS excluded). Each region receives the population-weighted MAP PfPR2-10 estimate for the survey year; regional all-under-five and neonatal mortality come from birth histories via the DHS synthetic-cohort life-table method over the 12 months before interview, and post-neonatal mortality is their difference. The assembled panel has 1,128 region-years (120 surveys, 36 countries); the primary sample requires country-mean PfPR2-10 above 1% and regional PfPR2-10 at least 1% (1,111 region-years), plus at least one post-neonatal and one neonatal death in the window, giving **1,055 region-years (118 surveys, 35 countries)**. The full data flow is shown in the analysis-flow diagram.

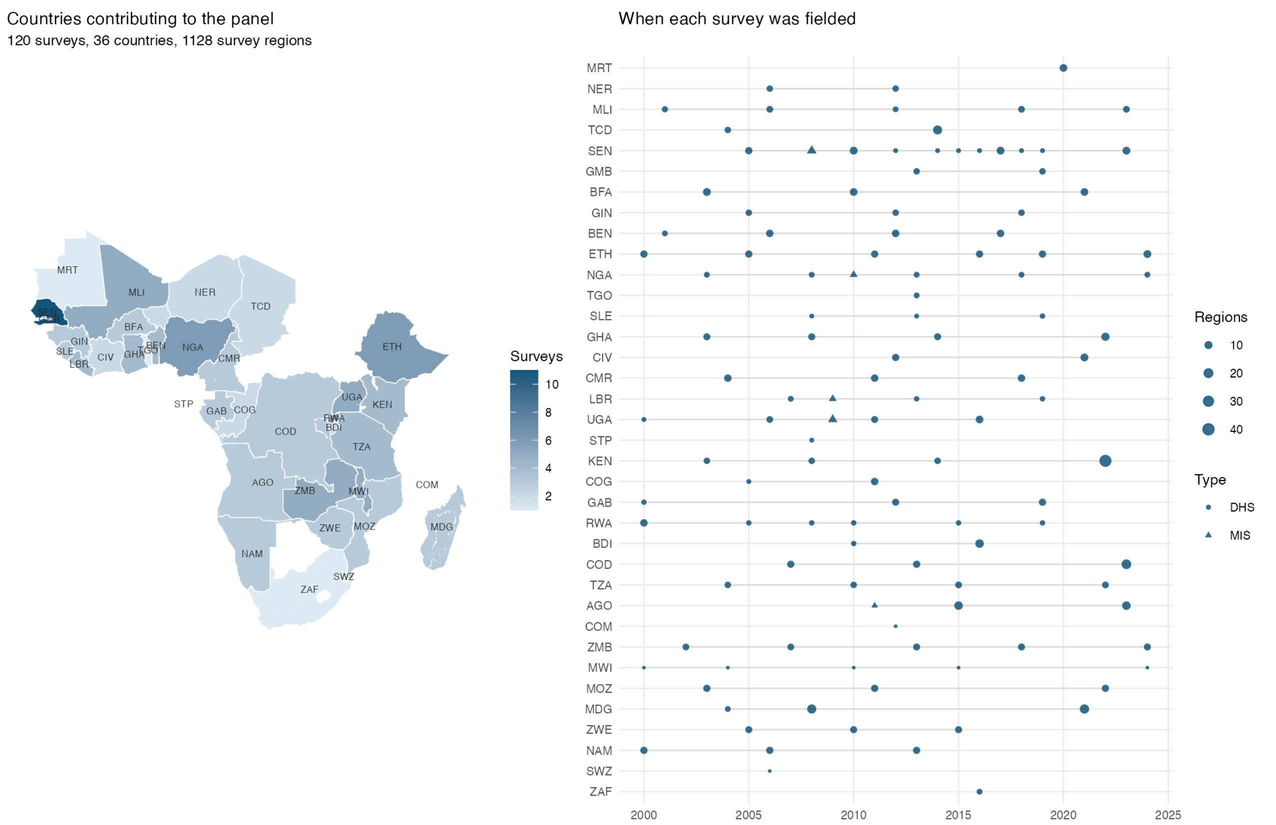

Figure 2: Where and when the surveys were fielded. Left: countries contributing to the panel, shaded by number of surveys. Right: fieldwork year of each survey, with point size giving the number of survey regions.

Mortality window. Regional mortality estimated over 12 months tracks the 60-month estimate closely for post-neonatal mortality (correlation 0.90 across 1,128 survey regions, median ratio 1.06) and less closely for neonatal mortality (correlation 0.65), with intervals about 1.9 times wider in both cases. The model comparison was repeated at 12, 24, 36, 48 and 60 months with MAP prevalence lagged 0, 1 or 2 years. The selected structure varies across the 15 cells (a time-interaction form in 12 of them). The attributable fraction at 30% prevalence ranges from 14% to 32% for the selected structure across the 15 cells and from 22% to 33% for the additive spline; at zero lag, the primary choice, the selected-structure range is 21% to 32%. The window changes precision far more than it changes the estimate; the prevalence lag matters more than the window.

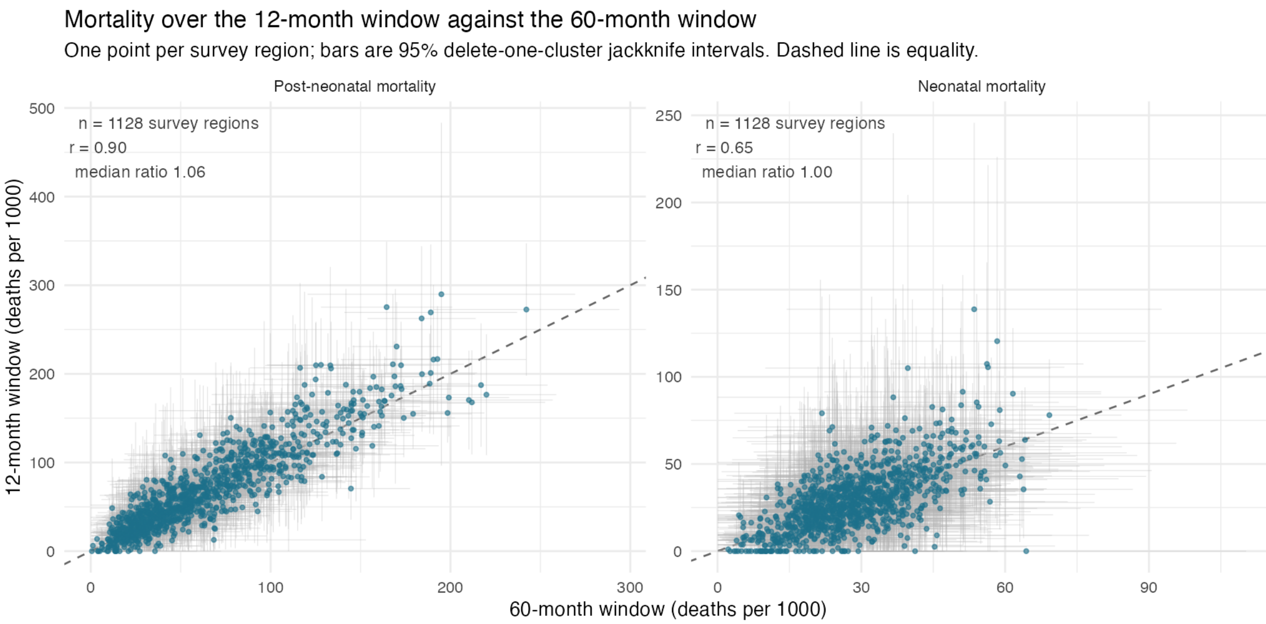

Figure 3: Post-neonatal and neonatal mortality over the 12-month window against the 60-month window, one point per survey region with 95% delete-one-cluster jackknife intervals.

Panel coverage. The panel is assembled from the DHS survey registry rather than inherited from an earlier aggregate. Three faults in that path were repaired — a Windows-1250 mis-decoding of accented region names, a `v022` that made `chmort` abort for five surveys (empty in four, and populated but with clusters not nested within strata in Malawi 2004), and a MAP regional extraction that had never completed — and recode region names are reconciled against boundary names across language, word order, spelling, qualifier clauses and granularity. Together these took the panel from 936 region-years (106 surveys) to 1,128 (120 surveys).

Of the 1,134 survey regions for which MAP publishes a prevalence estimate, **1,128 (99.5%) merge into the panel**. Only 995 of those joined on an exact key; the rest were reconciled by token signature (78), prefix (22), edit distance (18), a curated synonym table (7) or elimination (5).

This matters for interpretation, not only for coverage: on the 60-month window the linear no-interaction estimate fell at every step as coverage improved (+8.5%, +8.30%, +7.61%, +7.33% per 10 PfPR points across the four successive panels), which suggests the surveys previously dropped were not missing at random with respect to the association. Six regions across two surveys still cannot be joined — Mali 2012 did not survey four of its regions, and Uganda 2016's Buganda/Central labelling is genuinely ambiguous — and per-survey merge detail is recorded in `results/dhs_rebuild/region_merge_quality.csv`.

Covariates. Twenty-five candidate covariates are screened before imputation; those with at most 5% missingness are retained and singly imputed (country median, overall-median fallback). The **18 retained covariates** enter a single standardised, ridge-penalised block (proportions logit-transformed):

Level	Covariates
Survey-region (DHS)	urban residence, DTP3, measles, facility delivery, maternal education, exclusive breastfeeding, short birth interval, maternal age at first birth, improved water, improved sanitation, household electricity
National, exact year	WUENIC Hib3, PCV-completion, rotavirus-completion; and child (0–14) HIV prevalence (log)
National, nearest year	log GDP per capita, log health expenditure per capita, political stability

Seven candidates are excluded for excess missingness (the three direct DHS vaccine completion recodes, household wealth, and the three anthropometry measures). Political stability is read from the current World Bank `GOV_WGI_PV_EST` series after the `PV.EST` indicator was archived. National DTP3, electricity access and health-expenditure-share are dropped as redundant with their regional or per-capita counterparts.

Child HIV prevalence is derived from the UNAIDS 2025 estimates workbook (number of children 0–14 living with HIV) divided by the World Bank 0–14 population (`SP.POP.0014.TO`), joined by ISO3 and exact survey year, and entered on the log scale because it spans a very wide range across countries (country means from about 0.02% in Madagascar to about 3.8% in Eswatini). Nigeria and Comoros publish only a 15–19 series (4% of the sample) and are imputed with a provenance flag. Source choices are in `R_dhs/COVARIATE_SOURCES.md`.

Model. Approximate death counts (rate × birth exposure) are modelled with a negative-binomial GAM: `offset(log(exposure))`, the ridge covariate block and country random intercepts. Country-specific random PfPR slopes, present in earlier versions, are no longer in the primary model because they have no clear interpretation; re-admitting them is reported as a sensitivity. Five PfPR/time specifications are compared by ML/AIC on the same 1,055 observations, and the selected form is refitted by REML for post-neonatal, all-under-five and neonatal mortality.

Bayesian refits. The prevalence-by-time surface is also fitted in Stan via `brms`, with `t2(PfPR, year)` in place of `te()`, the ridge block as a `normal(0, 1)` prior on the standardised covariates, and half-t priors on the smoothing standard deviations, so that uncertainty in how smooth the surface is propagates into every attributable fraction rather than being fixed by an optimiser. The same programme is refitted to the neonatal outcome and to subgroups by era and region.

Association and model selection

From the linear summary models, each 10 percentage-point increase in PfPR2-10 is associated with:

Outcome	Change in mortality	95% CI	p-value
Post-neonatal	+6.92%	+4.52% to +9.37%	<0.001
All under-five	+4.67%	+2.77% to +6.61%	<0.001
Neonatal (negative control)	+0.12%	−2.18% to +2.48%	0.92

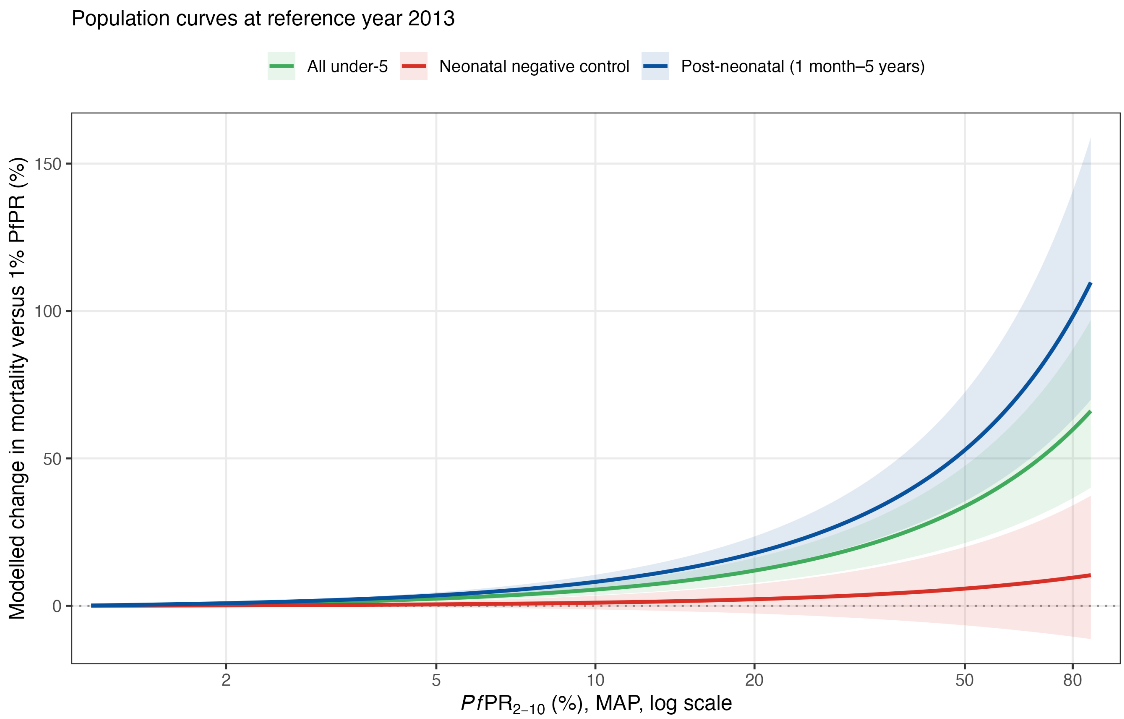

Figure 4: Modelled change in mortality versus 1% PfPR for the three outcomes. The post-neonatal and all-under-five associations rise with prevalence; the neonatal negative control stays flat and its interval spans zero.

Five specifications are compared by AIC on the 1,055-observation post-neonatal sample:

Model	Specification	AIC	ΔAIC
Linear, interaction	`PfPR + PfPR × year + s(year)`	8144.70	0.00
Spline, `ti` interaction	`s(PfPR) + s(year) + ti(PfPR, year)`	8144.80	0.11
Spline, no interaction	`s(PfPR) + s(year)`	8145.14	0.44
Full tensor surface	`te(PfPR, year)`	8150.71	6.01
Linear, no interaction	`PfPR + s(year)`	8151.06	6.37

The top three are indistinguishable by AIC. In the selected model the interaction is +0.0048 per year (`p=0.0003`): the proportional effect of prevalence is estimated to have strengthened over 2000–2024. An independent AIC comparison for neonatal mortality also lands on the linear time-interaction form (by 0.96 AIC over the no-interaction form), but with a null prevalence coefficient (+1.1% per 10 points, 95% CI −1.4% to +3.7%; `p=0.39`).

The selected specification gives the following malaria-attributable fractions of post-neonatal mortality (country random effects set to zero; reference year 2013):

PfPR2-10	Attributable fraction	95% interval
10%	7.5%	5.4%–9.5%
30%	22.2%	16.4%–27.6%
50%	34.6%	26.2%–42.0%

Does the prevalence effect change over time? A Bayesian comparison

Three models were fitted in `brms` with identical priors and differ only in how prevalence and calendar time enter: **A**, additive `s(PfPR) + s(year)`; **C**, A plus `s(PfPR, by = year)`, so the prevalence curve shifts linearly with time (pinned to zero at the 1% reference so it does not duplicate the linear year term); and **D**, the full `t2(PfPR, year)` surface. `brms` has no `ti()`, so D is not nested above A or C and the comparison is predictive rather than a test of one term.

Model	Pointwise LOO elpd	Survey-grouped 10-fold elpd	Difference from best (SE)	Stacking weight (10-fold)
D: full surface	−4072.3	−4129.6	0.0	0.86
C: curve shifts linearly in time	−4072.1	−4130.6	−1.1 (1.7)	0.14
A: additive	−4072.4	−4136.7	−7.1 (3.4)	0.00

Pointwise leave-one-out cannot separate the three: the other regions of the same survey stay in the training set, so the country intercept and the year smooth already pin down that survey's level. Holding out whole surveys (10 folds of 11–12 surveys, identical across models) asks whether the time structure generalises to a survey the model has never seen, and there the additive model falls behind by 7.1 (about two pointwise standard errors; 1.6 when the 118 surveys rather than the 1,055 region-years are treated as the exchangeable units). The gain is consistent (70 of 118 surveys favour D; removing the three most favourable still leaves 3.3) and sits almost entirely in survey regions with prevalence of 30% or more (+6.7, SE 2.4) and in West Africa (+5.5, SE 2.0). C and D are indistinguishable.

Under C and D the attributable fraction at 50% prevalence is higher in 2020 than in 2005 by 17 and 23 percentage points respectively (95% intervals 2–32 and 6–40); at 30% the intervals include zero (11, −3 to 24; 15, −1 to 31). The interaction is a real feature of the posterior, but it lives at high prevalence in recent years, where the data are thinnest.

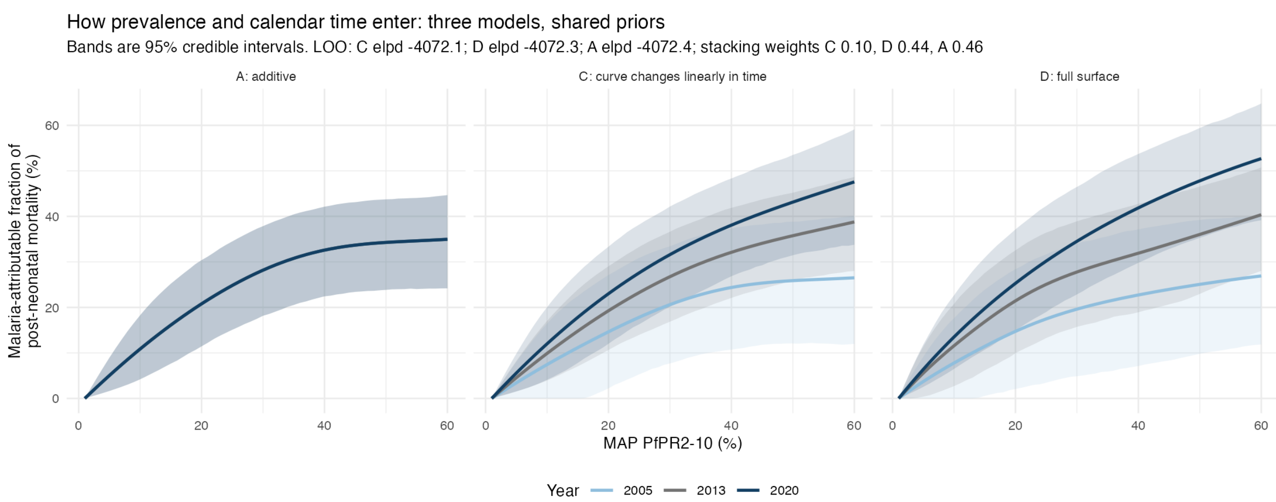

Figure 5: The three models with shared priors, curves at 2005, 2013 and 2020 with 95% credible intervals. In A the three years coincide by construction.

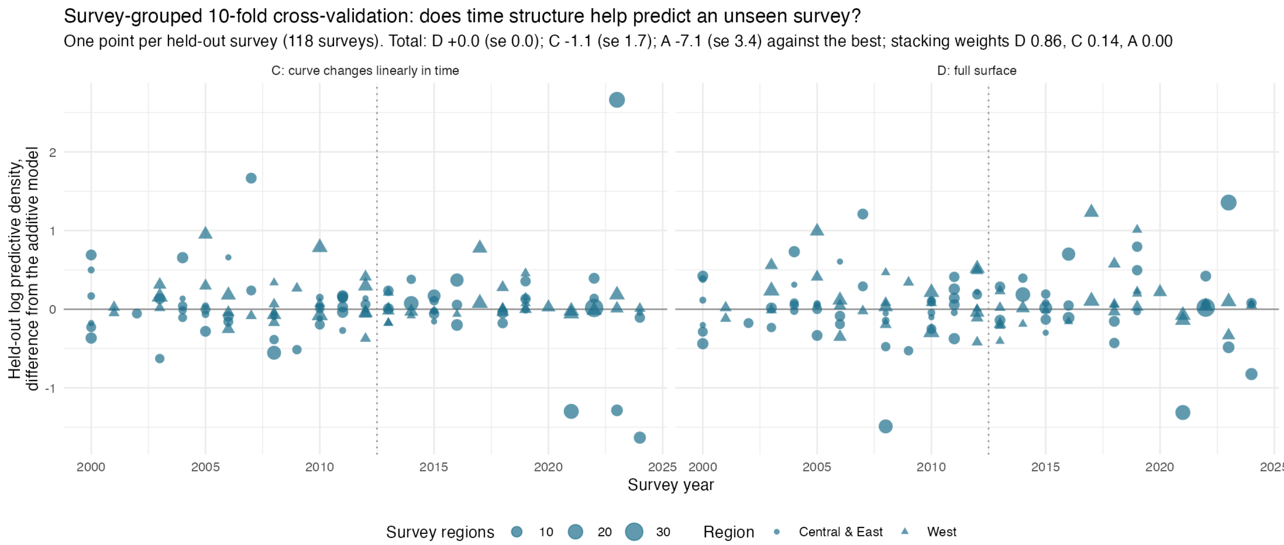

Figure 6: Survey-grouped 10-fold cross-validation: held-out log predictive density per survey, relative to the additive model. Points above zero favour the time-structured model.

Subgroups by era and region

The same Bayesian structure fitted separately by era (before/from 2013, the median survey year) and region (West Africa versus Central and East) gives the following attributable fractions at 30% prevalence (95% credible intervals; n is surveys):

Subgroup	Surveys	AF at 30% PfPR
All data	118	28.3% (17.7%–37.6%)
Before 2013	66	19.9% (6.4%–34.5%)
2013 onwards	52	35.3% (21.5%–47.8%)
West Africa	51	31.8% (11.7%–49.8%)
Central and East	67	21.3% (7.0%–34.0%)
West, before 2013	25	47.2% (16.1%–66.9%)
Central and East, before 2013	41	8.0% (0.0%–24.8%)
West, 2013 onwards	26	20.3% (0.0%–45.9%)
Central and East, 2013 onwards	26	38.3% (22.5%–52.5%)

The era contrast is the clear signal and is consistent with the time interaction above. The regional contrast is not: the credible intervals overlap along the whole prevalence range, a three-way model with region-specific surfaces buys nothing in leave-one-out (elpd difference 0.3, SE 2.7), and the 2×2 cells occupy different prevalence ranges, so the apparent crossover between West and Central–East across eras should not be over-read. The neonatal control fitted the same way straddles zero in every subgroup.

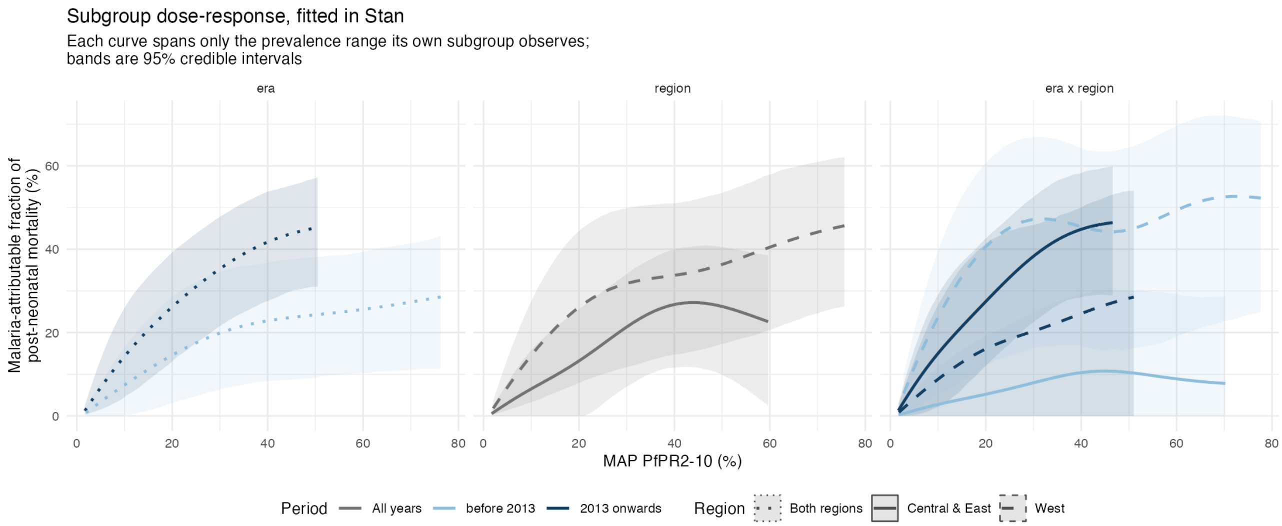

Figure 7: Subgroup dose-response curves, fitted in Stan. Dashed lines are West Africa, solid Central and East; light colours are before 2013, dark from 2013. Each curve spans only the prevalence range its subgroup observes.

Robustness

Robustness is the central message: the association barely moves across covariate blocks, samples, model structures and likelihood.

- **Covariate adjustment.** Child HIV prevalence is associated with +8.2% higher post-neonatal mortality per standard deviation (`p=0.012`), behind regional DTP3 coverage (−9.5%) and maternal age at first birth (−7.6%), yet adding the national block leaves the malaria estimate essentially unchanged: the regional block alone and the full block give attributable fractions of 24.0% and 22.3% at 30% PfPR.

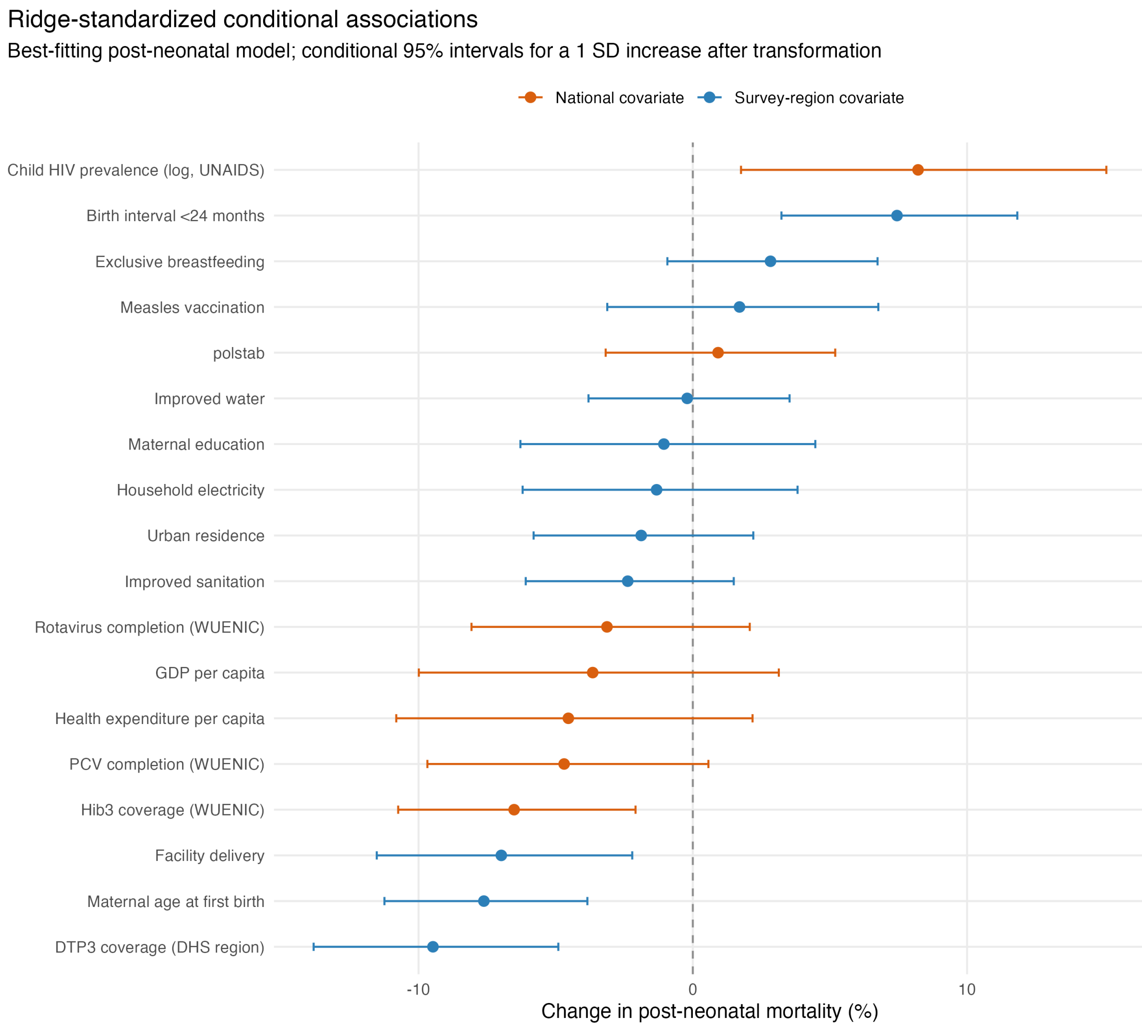

Figure 8: Ridge-standardised conditional associations for the 18 covariates in the selected post-neonatal model (percentage change per 1 SD after transformation; ridge-shrunk, not independent causal effects).

- **Samples, structure and likelihood.** The attributable fraction at 30% PfPR stays near 22% across alternative samples (22.2% including sub-1% regions, 23.6% on complete cases) and model-structure choices (23.3% with country-specific PfPR slopes re-admitted, 24.0% with regional covariates only); it rises on the 5–40% prevalence range (30.0%, a smaller sample of 637) and in a national-covariates-only model that fits far worse (31.0%, AIC +176). A log-Gaussian rate model gives +5.6% per 10 PfPR points (95% CI +0.4% to +10.9%; `p=0.034`), less precise than the count model but now excluding zero.
- **Country random effects.** Re-admitting the country-specific PfPR slope moves the national burden by at most 7.7% in any year, and including each in-sample country's own random effects rather than the population average moves it by at most 7.1% in the other direction.

External validation: bednet trials

An independent triangulation against eight insecticide-treated-net/curtain cluster-RCTs — which measured both parasite prevalence and child mortality — gives a slope of the same order as the observational model: a weighted meta-regression gives about an 8.6% mortality reduction per 10-point drop in age-standardised PfPR2-10, against the model's linear 6.9% (the trial estimate is imprecise — `p=0.12`, eight heterogeneous trials — and is wholly external to the panel). Predicted and observed trial effects broadly track each other.

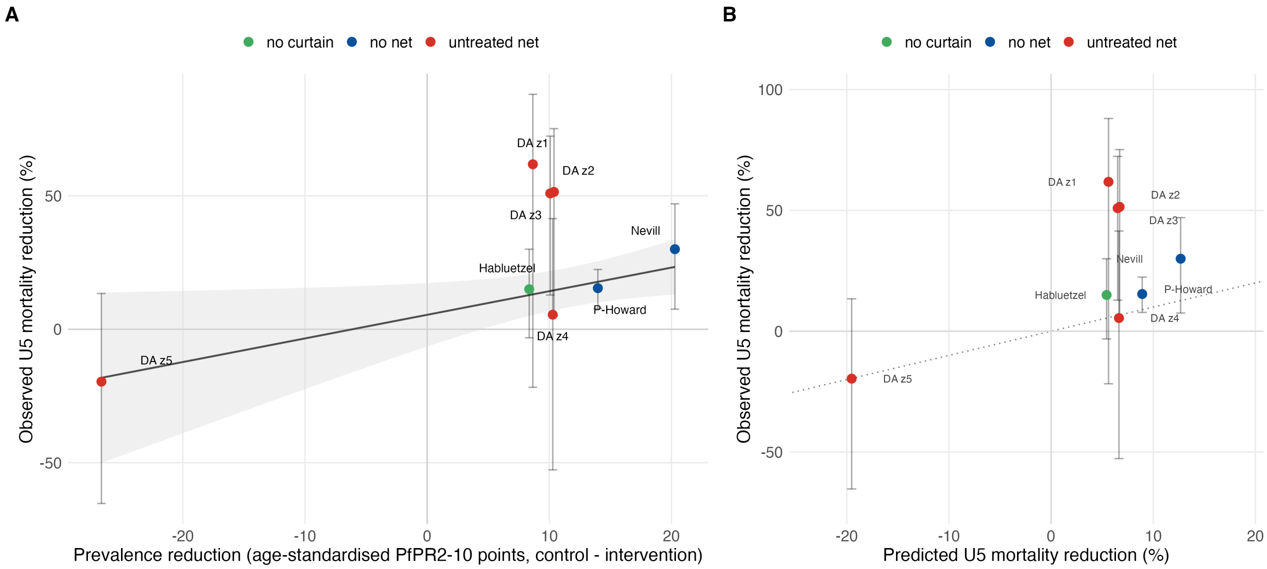

Figure 9: RCT triangulation. **A:** observed under-five mortality reduction versus the age-standardised prevalence reduction across trial arms (weighted fit). **B:** mortality reductions predicted from the model's prevalence slope versus observed (dotted line = 1:1).

National burden

Latest-year comparison

The comparison is not perfectly contemporaneous: the model surface is evaluated at 2024 using the latest available 2024 MAP PfPR, IGME all-cause mortality and live births for 43 matched sub-Saharan African countries; IHME is 2024 under-five malaria deaths; WHO WMR 2025 reports all-age country deaths for 2024, so the under-five comparator applies WHO's ~75% age-share statement. Three time structures are extrapolated; the spline with a `ti` interaction fits best of the three (AIC 8144.8, against 8145.1 for the time-stable spline and 8150.7 for the `te` surface).

Source	Deaths	95% interval
Spline, `ti` interaction (best of three)	**527,000**	373,000–658,000
Full `te` surface	503,000	304,000–666,000
Spline, no time interaction	435,000	291,000–565,000
IHME/GBD SSA under-five, 2024	428,000	392,000–469,000
WHO African-region all-age, 2024	579,000	531,000–706,000
WHO African-region under-five proxy, 2024	434,000	398,000–530,000

The model intervals propagate both the fitted attributable-fraction uncertainty and the all-cause post-neonatal mortality uncertainty (from the IHME/GBD relative interval).

Temporal trend

The historical trend is the quantity most sensitive to whether a time interaction is admitted. Applied to national PfPR and all-cause mortality series, estimated post-neonatal malaria deaths from 2000 to 2024 change by:

Model	2000	2024	Change
Spline `ti` interaction (best of three)	553,000	527,000	−5%
Full `te` surface	486,000	503,000	+4%
Spline, no time interaction	925,000	435,000	−53%

The spread is the headline uncertainty in the trend, and it is wide: a time-stable attributable fraction implies the burden more than halved over the period, whereas the time-varying structures, which let the attributable fraction rise as prevalence fell, imply little change. The two best structures are 0.3 AIC apart. The survey-grouped cross-validation above favours the time-varying structures over the additive one, but modestly, and mainly through high-prevalence regions. The trend should be read as a range, not a point.

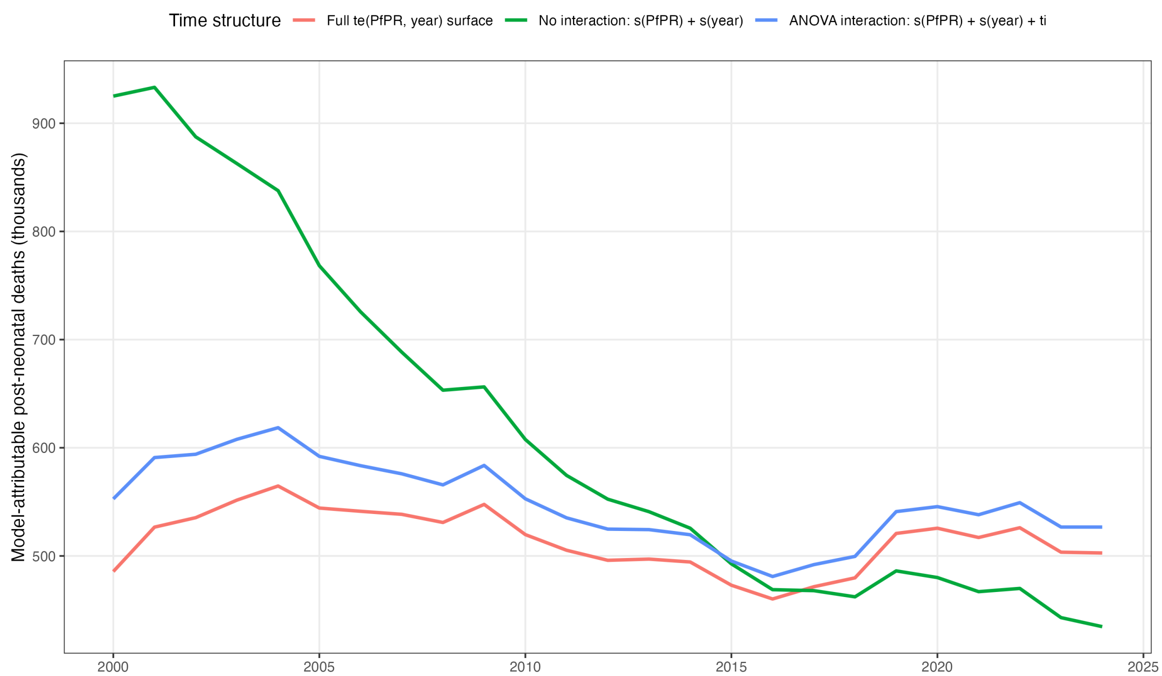

Figure 10: National malaria-attributable post-neonatal deaths over time under the three time structures.

Country differences from IHME

The aggregate is about 23% above IHME (model 527,000 vs IHME 428,000 in 2024, a ratio of 1.23), and the country-level differences are larger in both directions (spline `ti` model; ten countries with the largest absolute difference from IHME/GBD, all for 2024):

Country	Model	IHME 2024	Model/IHME
DR Congo	125,600	61,800	2.03
Nigeria	192,100	138,100	1.39
Uganda	14,300	26,100	0.55
Burundi	4,200	14,400	0.29
Chad	13,000	4,900	2.67
Ethiopia	4,300	8,900	0.48
Tanzania	5,400	9,900	0.55
Mozambique	14,200	10,300	1.38
Cameroon	13,100	16,300	0.80
Ghana	3,800	6,700	0.56

The model sits well above IHME in DR Congo, Nigeria and Chad and well below it in Uganda, Burundi, Ethiopia and Tanzania. DR Congo alone accounts for about 64,000 of the 99,000 aggregate excess. (IHME/GBD malaria estimates are unavailable for Sudan and South Africa, so the IHME total covers 41 of the 43 countries; both contribute negligibly.)

Interpretation and limitations

The data provide fairly robust evidence of a positive cross-sectional association between malaria prevalence and post-neonatal mortality that survives adjustment for national vaccine rollout and child HIV prevalence, the neonatal negative control, alternative samples, model-structure choices and likelihood. The main remaining limitations are:

- the aggregate burden is not a validation against a common estimand — the model includes indirect malaria-associated deaths, whereas IHME and WHO assign deaths specifically to malaria, and national extrapolation uses population-average effects that ignore country random effects;
- the 12-month window buys alignment between outcome and exposure at the price of precision, and 56 region-years with no death of one type in the window are excluded from the shared sample;
- the historical trend is not well identified. Three specifications are separated by 0.44 AIC yet imply changes from −53% to +4% over 2000–2024, and the evidence for a time-varying effect, while consistent across held-out surveys, rests on high-prevalence regions in recent years where data are thin. The trend should be read as a range, not a point;
- 2025 exposure and all-cause mortality inputs are unavailable, and the WHO under-five comparison applies a regional 75% age-share to every country;
- child HIV prevalence is derived from UNAIDS 0–14 counts and a World Bank denominator rather than a published rate, and Nigeria and Comoros (about 4% of the sample, including the largest-burden country) lack an under-15 series and are imputed;
- the Bayesian fits fix the ridge penalty on the covariate block at its standardised scale rather than estimating it, and their smoothing parameters are weakly identified (posterior close to prior), which is why their intervals are wider than the `mgcv` ones;
- on the 60-month window the dose-response estimate fell monotonically as survey coverage was repaired (+8.5% to +7.3% per 10 PfPR points across four successive panels). That pattern is not noise-shaped and suggests the previously dropped surveys differed systematically; further coverage work could move it again.

Changes from the 28 August version. The mortality window moved from 60 to 12 months (sample 1,111 to 1,055); country-specific random PfPR slopes were removed from the primary model; the neonatal control moved from marginally positive to null; the AIC selection became a three-way tie rather than a clear preference for the time interaction; Bayesian refits, a survey-grouped cross-validation and era-by-region subgroups were added; the 2024 burden moved from 546,000 to 527,000 and the country pattern against IHME is again mixed rather than one-directional.

Analysis flow

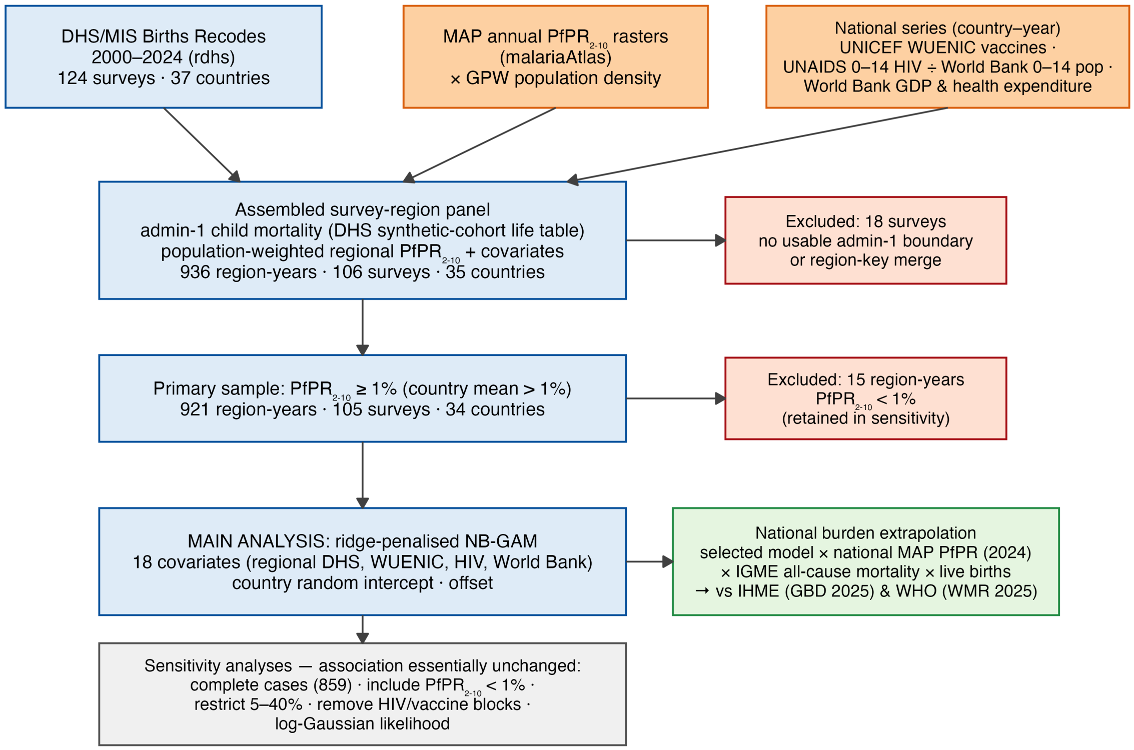

Figure 11: Data and analysis flow. External data (MAP PfPR2-10 rasters, UNICEF WUENIC vaccine coverage, UNAIDS/World Bank child HIV prevalence, and World Bank economic series) feed the assembled survey-region panel, which is filtered to the primary sample, analysed with the ridge-penalised model, and extrapolated to a national burden.